

OS Unit-4 Banana

PES2UG24AM009

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Github Link-

<https://github.com/umbralrage345/PES2UG24AM009-PES-VCS.git>

Output Screenshots:

Phase A:

1.A

```
abhyudaya@Abhyudaya: /mn  x  +  v
In file included from /usr/include/stdio.h:980,
                 from pes.h:8,
                 from object.c:11:
In function 'snprintf',
    inlined from 'object_write' at object.c:132:5:
/usr/include/x86_64-linux-gnu/bits/stdio2.h:54:10: note: '__builtin___snprintf_chk' output between 5 and 516 bytes into a destination of size 512
54 |     return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEVEL - 1,
    |
55 |                                     __glibc_objsize (__s), __fmt,
    |
56 |                                     __va_arg_pack ());
    |
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5db0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5db0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5db0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$
```

1.B

```

abhyudaya@Abhyudaya:/mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aaafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aaafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
abhyudaya@Abhyudaya:/mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$

```

Phase 2-

2.A

```

abhyudaya@Abhyudaya:/mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
abhyudaya@Abhyudaya:/mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$

```

2.B

```

abhyudaya@Abhyudaya:/mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$ xxd .pes/objects/2c/7bfbcb68641e9c8385cca074e2c2558482decda005cb883e0b08
f08afe571d | head -20
00000000: 7472 6565 2039 3800 3130 3037 3535 2066  tree 98.100755 f
00000010: 696c 6531 2e74 7874 002c f8d8 3d9e e295  ilel.txt,...=...
00000020: 43b3 4a87 7274 21fd ecb7 e3f3 a183 d337  C.J.rt!.....7
00000030: 6390 25de 576d b9eb b431 3030 3735 3520  c.%Wm...100755
00000040: 6669 6c65 322e 7478 7400 e00c 50e1 6a2d  file2.txt..P.j-
00000050: f38f 8d6b f809 e181 ad02 48da 6e67 19f3  ...k.....H.ng..
00000060: 5f9f 7e65 d6f6 0619 9f7f                                     ~-@.....
abhyudaya@Abhyudaya:/mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$

```

Phase 3

3.A

```
abhyudaya@Abhyudaya: /mn x + v
|
index.c: At top level:
index.c:180:12: warning: 'compare_index_entries' defined but not used [-Wunused-function]
180 | static int compare_index_entries(const void *a, const void *b) {
    |
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$ rm -rf .pes
./pes init
echo hello > file1.txt
echo world > file2.txt
./pes add file1.txt file2.txt
./pes status
cat .pes/index
Initialized empty PES repository in .pes/
Staged changes:
staged:    file1.txt
staged:    file2.txt

Unstaged changes:
(nothing to show)

Untracked files:
untracked: .gitignore
untracked: commit.c
untracked: commit.h
untracked: index.c
untracked: index.h
untracked: Makefile
untracked: object.c
untracked: pes.c
untracked: pes.h
untracked: README.md
untracked: test_objects.c
untracked: test_sequence.sh
untracked: test_tree.c
untracked: tree.c
untracked: tree.h

100755 2cf8d83d9ee29543b34a87727421fdceb7e3f3a183d337639025de576db9ebb4 1776468241 6 file1.txt
100755 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776468241 6 file2.txt
abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$
```

3.B

```
abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$ xxd .pes/objects/2c/7bfb6c68641e9c8385cca074e2c2558482decde005cb883e60b08
f08afe571d | head -20
00000000: 7472 6565 2039 3800 3130 3037 3535 2066  tree 98.100755 f
00000010: 696c 6531 2e74 7874 002c f8d8 3d9e e295  ile1.txt,...=...
00000020: 43b2 4a87 7274 21fd ecb7 e3f3 a183 d337  C.J.rtl.....7
00000030: 6390 25de 576d b9eb b431 3030 3735 3520  c.%.wm...100755
00000040: 6669 6c65 322e 7478 7400 e00c 50e1 6a2d  file2.txt...P.j-
00000050: f38f 8d6b f809 e181 ad02 48da 6e67 19f3  ...k.....H.ng..
00000060: 5f9f 7e65 d6f6 0619 0f7f  -==e.....
abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$ cat .pes/index
100755 2cf8d83d9ee29543b34a87727421fdceb7e3f3a183d337639025de576db9ebb4 1776468888 6 file1.txt
100755 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776468888 6 file2.txt
abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$
```

Phase 4

4.A

```
abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$ cat .pes/index
100755 2c8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 1776468888 6 file1.txt
100755 e08c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776468888 6 file2.txt
abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$ rm -rf .pes
./pes init

echo one > file1.txt
./pes add file1.txt
./pes commit -m "commit one"

echo two >> file1.txt
./pes add file1.txt
./pes commit -m "commit two"

echo three >> file1.txt
./pes add file1.txt
./pes commit -m "commit three"

./pes log
Initialized empty PES repository in .pes/
Committed: fc27e09aa5ba... commit one
Committed: 32804badbd06... commit two
Committed: 1b410fd99c54... commit three
commit 1b410fd99c54ef098f7a2685ef7ebca3f23fballe4a38edd4807d8b79838ef4a
Author: PES User <pes@localhost>
Date: 1776472020

    commit three

commit 32804badbd069cc92425b2d8fe3ab6730e8b9f1e8038e58c401573c54395d81e
Author: PES User <pes@localhost>
Date: 1776472020

    commit twoee

commit fc27e09aa5ba63c8bf0e439f7f5403c48bb7cd9ed1bad3d970ada2c658f7723e
Author: PES User <pes@localhost>
Date: 1776472020

    commit one
commit oner <Q

abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$
```

4.B

```
abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/1b/410fd99c54ef098f7a2685ef7ebca3f23fballe4a38edd4807d8b79838ef4a
.pes/objects/32/804badbd069cc92425b2d8fe3ab6730e8b9f1e8038e58c401573c54395d81e
.pes/objects/47/2537b7bbcbfb122637254956d7212d25447704d3786b7170093a7344296c17
.pes/objects/71/809267f426a1fbc1685dacb77f6192526e9739dbafec3ac76d1e094c5a4ef6
.pes/objects/82/8e04ddaf383457e63ca44774c298b2d56b4339d47fdfa929d8bc0b2922410f
.pes/objects/a4/ed1f355af02d88cd291d0e4463910c5061ec48a49aa2b1539b9af973b286
.pes/objects/a6/b74238e52ca07a0ce235197f8b444a58b98ad8a0e07a20a896e15345546da8
.pes/objects/c5/df06a7d3510bf59e7cd9ac36db54a7c0b3eb45141eab0d9da7f66c50f96e77
.pes/objects/fc/27e09aa5ba63c8bf0e439f7f5403c48bb7cd9ed1bad3d970ada2c658f7723e
.pes/refs/heads/main
abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$
```

4.C

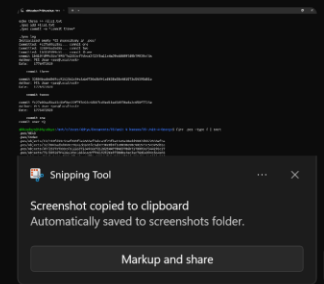
```
abhyudaya@Abhyudaya: /mn
./pes log
Initialized empty PES repository in .pes/
Committed: fc27e09aa5ba... commit one
Committed: 32804badbd06... commit two
Committed: 1b410fd99c54... commit three
commit 1b410fd99c54ef098f7a2685ef7ebca3f23fballe4a38edd4807d8b79838ef4a
Author: PES User <pes@localhost>
Date: 1776472020

commit three
commit 32804badbd069cc92425b2d8fe3ab6730e8b9f1e8038e58c401573c54395d81e
Author: PES User <pes@localhost>
Date: 1776472020

commit twoee
commit fc27e09aa5ba63c8bf0e439f7f5403c48bb7cd9ed1bad3d970ada2c658f7723e
Author: PES User <pes@localhost>
Date: 1776472020

commit one
commit oner <Q

abhyudaya@Abhyudaya:/mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/1b/410fd99c54ef098f7a2685ef7ebca3f23fballe4a38edd4807d8b79838ef4a
.pes/objects/32/804badbd069cc92425b2d8fe3ab6730e8b9f1e8038e58c401573c54395d81e
.pes/objects/47/2537b7bbcbfb122637254956d7212d25447704d3786b7170093a7344296c17
.pes/objects/71/809267f426a1fbc1685dacb77f6192526e9730dbafec3ac76d1e094c5a4ef6
.pes/objects/82/8e04ddaf383457e63ca44774c298b2d56b4339d47fdfa929d8bc0b2922410f
.pes/objects/a4/ed1f355af02d88cd291d0e4463910c5061ece48a49aa2b1539b9af973b286
.pes/objects/a6/b74238e52ca07a0ce235197f8b444a58b98ad8a0e07a20a896e15345546da8
.pes/objects/c5/df06ad3d510bf59e7cd9ac36db54a7c0b3eb45141eab0d9da7f66c50f96e77
.pes/objects/fc/27e09aa5ba63c8bf0e439f7f5403c48bb7cd9ed1bad3d970ada2c658f7723e
.pes/refs/heads/main
abhyudaya@Abhyudaya:/mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$ cat .pes/refs/heads/main
cat .pes/HEAD
1b410fd99c54ef098f7a2685ef7ebca3f23fballe4a38edd4807d8b79838ef4a
ref: refs/heads/main
abhyudaya@Abhyudaya:/mnt/c/Users/abhyu/Documents/OS/unit 4 banana/OS-Unit-4-Orange$
```



Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

Full integration test

```
abhyudaya@Abhyudaya: /mn x + v
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:    file.txt
  staged:    hello.txt

Unstaged changes:
(nothing to show)

Untracked files:
(nothing to show)

--- First Commit ---
Committed: 575f36f641e0... Initial commit

Log after first commit:
commit 575f36f641e0df1e01bb34101920c1851cf441bd1bcd86e5c4c8b7e9add0d6a2
Author: PES User <pes@localhost>
Date:   1776472174

    Initial commit

--- Second Commit ---
Committed: 86738a72bbe0... Update file.txt

--- Third Commit ---
Committed: 6d7b11e3f158... Add farewell

--- Full History ---
commit 6d7b11e3f158b7e876ae973db623c74c15bfdcd0e8bce415563c49d66bd7c031
Author: PES User <pes@localhost>
Date:   1776472174

    Add farewell

commit 86738a72bbe0b082f544e14c5073cc00675c66c0e0a1a7fa96b2d37cdceda529
Author: PES User <pes@localhost>
Date:   1776472174

    Update file.txt

commit 575f36f641e0df1e01bb34101920c1851cf441bd1bcd86e5c4c8b7e9add0d6a2
Author: PES User <pes@localhost>
Date:   1776472174

    Initial committial commiQ

--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
6d7b11e3f158b7e876ae973db623c74c15bfdcd0e8bce415563c49d66bd7c031

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/57/5f36f641e0df1e01bb34101920c1851cf441bd1bcd86e5c4c8b7e9add0d6a2
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/6d/7b11e3f158b7e876ae973db623c74c15bfdcd0e8bce415563c49d66bd7c031
.pes/objects/86/738a72bbe0b082f544e14c5073cc00675c66c0e0a1a7fa96b2d37cdceda529
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef9e08f7fa1974d901ce282f30e08decccd92
.pes/objects/d0/7733c25d2d17b7574be8c5502b562bf48bafaa3029f61f75b8010d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6cf

=== All integration tests completed ===
make: warning: Clock skew detected. Your build may be incomplete.
abhyudaya@Abhyudaya: /mnt/c/Users/abhyu/Documents/PESFinal$
```

Phase 5: Branching and Checkout

Q5.1 How would you implement pes checkout <branch>? What files need to change in .pes/, and what must happen to the working directory? What makes this operation complex?

To implement pes checkout <branch>, the system must first verify that the requested branch exists inside .pes/refs/heads/. Each branch is stored as a file containing the latest commit hash of that branch.

The following steps would be performed:

Read the target branch file from .pes/refs/heads/<branch>.

Obtain the commit hash stored in that file.
Load the commit object from the object store.
Read the tree object referenced by that commit.
Compare the current working directory with the target tree.
Delete files that exist in the current branch but not in the target branch.
Create or overwrite files that exist in the target branch.
Update .pes/HEAD so it points to:
ref: refs/heads/<branch>
Update the staging index .pes/index so it matches the checked-out branch.
Files changed inside .pes/:
.pes/HEAD
.pes/index
The working directory must be rewritten to exactly match the files stored in the selected branch snapshot.
This operation is complex because:
Files may differ between branches.
Some files may need deletion.
Some files may need replacement.
Local uncommitted changes can be lost.
Directory creation/removal must be handled carefully.
Binary files and permissions must be restored properly.

Q5.2 How would checkout detect a dirty working directory conflict using only the index and object store?

A dirty working directory means tracked files were modified after staging or after the last commit.

To detect conflict:

Read current tracked files from .pes/index.

For each tracked file:

Check if file exists.

Compare file size and modification time with index values.

Recalculate hash if needed.

If file content differs from staged hash, it is modified.

Then compare with target branch tree:

If the same file would also be replaced during checkout, conflict exists.

Example:

Current branch:

file.txt = Hello

User modifies:

file.txt = My changes

Target branch:

file.txt = Different version

Checkout must refuse because user changes would be overwritten.

System should print:

error: local changes would be overwritten by checkout

Q5.3 What is Detached HEAD? What happens if commits are made there? How can user recover them?

Detached HEAD means .pes/HEAD stores a commit hash directly instead of pointing to a branch reference.

Example:

6df7b11e...

instead of:

ref: refs/heads/main

In this state:

HEAD points directly to one commit.

New commits can still be created.

But no branch name moves forward.

So commits become unreachable later if user switches branches.

Recovery methods:

Create a branch at current commit:

pes branch rescue

pes checkout rescue

Or manually save the commit hash in a branch reference file.

This preserves detached commits from garbage collection.

Phase 6: Garbage Collection and Space Reclamation

Q6.1 Describe algorithm to find and delete unreachable objects. What data structure is used?

Estimate visits for 100,000 commits and 50 branches.

Garbage collection removes objects not reachable from any branch.

Reachable objects include:

Commit objects

Tree objects

Blob objects

Algorithm:

Step 1: Start from all branch heads

Read every file inside:

.pes/refs/heads/

Suppose 50 branches.

Step 2: Graph Traversal

For each branch commit:

Mark commit reachable.

Visit parent commit.

Visit tree object.

Visit blobs inside tree.

Use DFS or BFS.

Data Structure

Use:

Hash Set for visited object hashes

Stack or Queue for traversal

Step 3: Scan object store

Check every object in:

.pes/objects/

If object hash not in reachable set, delete it.

Complexity Estimate

If repository has:

100,000 commits

50 branches

Many branches share history.

Actual unique commit visits $\approx$ 100,000 not 5,000,000.

Then plus trees/blobs.

Approximate total visits:

100,000 commits

100,000 trees

Several thousand blobs

Total around 200,000 to 500,000 object visits depending on project size.

Q6.2 Why is concurrent garbage collection dangerous? Explain race condition. How does real Git avoid it?

Garbage collection running during commit creation is dangerous.

Race Condition Example

Commit process writes new blob object.

Blob exists in .git/objects/

Commit has not yet updated branch reference.

GC scans reachable objects.

New blob is not yet referenced.

GC deletes blob.

Commit later tries to reference deleted blob.

Result:

Corrupted commit

Missing object errors

How Real Git Avoids This

Git prevents this using:

Lock files

Example:

refs/heads/main.lock

Atomic rename operations

Temporary object protection

Reflogs to preserve recent references

GC usually avoids deleting very recent loose objects.

Thus objects being written are protected until references are safely updated.